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Draft Leaving Certificate Mathematics Specification Higher/Ordinary Level

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Senior cycle

Senior cycle aims to educate the whole person and contribute to human flourishing. Students' experiences throughout senior cycle enrich their intellectual, social and personal development and their overall health and wellbeing. Senior cycle has 8 guiding principles.

Senior Cycle Guiding Principles	
Wellbeing and relationships	Choice and flexibility
Inclusive education and diversity	Continuity and transitions
Challenge, engagement and creativity	Participation and citizenship
Learning to learn, learning for life	Learning environments and partnerships

These principles are a touchstone for schools and other educational settings, as they design their senior cycle. Senior cycle consists of an optional Transition Year, followed by a two-year course of subjects and modules. Building on junior cycle, learning happens in schools, communities, educational settings and other sites, where students' increasing independence is recognised. Relationships with teachers are established on a more mature footing and students take more responsibility for their learning.

Senior cycle provides a curriculum which challenges students to aim for the highest level of educational achievement, commensurate with their individual aptitudes and abilities. During senior cycle, students have opportunities to grapple with social, environmental, economic and technological challenges and to deepen their understanding of human rights, social justice, equity, diversity and sustainability. Students are supported to make informed choices as they choose different pathways through senior cycle and every student has opportunities to experience the joy and satisfaction of reaching significant milestones in their education. Senior cycle should establish firm foundations for students to transition to further, adult and higher education, apprenticeships, traineeships and employment and participate meaningfully in society, the economy and adult life.

The educational experience in senior cycle should be inclusive of every student, respond to their learning strengths and needs and celebrate, value and respect diversity. Students vary in their family and cultural backgrounds, languages, age, ethnic status, beliefs, gender and sexual identity as well as their strengths, needs, interests, aptitudes and prior knowledge, skills, values and dispositions. Every student's identity should be celebrated, respected and responded to throughout their time in senior cycle.

At a practical level, senior cycle is supported by enhanced professional development; the involvement of teachers, students, parents, school leaders and other stakeholders; resources; research; clear communication; policy coherence; and a shared vision of what senior cycle seeks to achieve for our young people as they prepare to embark on their adult lives. It is brought to life in schools and other educational settings through:

- effective curriculum planning, development, organisation, reflection and evaluation
- teaching and learning approaches that motivate students and enable them to improve
- a school culture that respects students and promotes a love of learning.

Rationale

Leaving Certificate Mathematics provides students with opportunities to see how mathematical thinking can help them understand and influence the world around them. Students learn that mathematics is an intellectual and aesthetic discipline which is continuously evolving. Its significance permeates all aspects of life, from the patterns and relationships found in nature, to the reasoning and analysis that guide decision-making, to the visual-spatial thinking and the creativity driving modern advancements. In a global context, mathematical understanding has become increasingly vital for addressing challenges such as cyber security, climate change, biodiversity loss and health-related issues. Students also come to appreciate that the study of mathematics for its own sake can be a very rewarding pursuit.

Mathematics in senior cycle offers continuity and progression from junior cycle. It is underpinned by the understanding of mathematics as an interconnected body of ideas and reasoning processes, developed collaboratively with teachers and peers as well as independently. The specification emphasises the interconnected nature of mathematics and the importance of developing mathematical proficiency. It also acknowledges the need to support students to be able to make informed judgments in their personal and civic life in the increasingly data-rich contexts of contemporary life. The specification recognises the role of digital technologies in enhancing learning, supporting visualisation and enabling richer engagement with complex ideas.

The study of mathematics develops a wide array of flexible and transferable learning about problem-solving, logical thinking, communication, creativity, collaboration, time management and decision-making. Improving mathematical proficiency, particularly adaptive reasoning and

strategic competence, helps students develop flexibility, challenge assumptions and cultivate rigour and perseverance. This learning can be applied to further study, life and work.

Aims

Leaving Certificate Mathematics aims to develop mathematical knowledge, skills, values and dispositions needed for continuing education, life and work. By teaching mathematics in contexts that help students make connections across mathematics itself, with other subjects and with real-life applications, it is envisaged that students will experience mathematics as an accessible, enjoyable and valuable field of study. In doing so, students develop a flexible, disciplined approach to thinking, an ability to engage thoughtfully with data and statistical information, and the confidence and enthusiasm to search for creative solutions to problems across a range of contexts.

More specifically, Leaving Certificate Mathematics aims to empower students to develop mathematical proficiency, characterised as

- **conceptual understanding** – comprehension of mathematical concepts, operations and relations
- **procedural fluency** – skill in carrying out procedures flexibly, accurately, efficiently and appropriately
- **strategic competence** – ability to formulate, represent and solve mathematical problems in both familiar and unfamiliar contexts
- **adaptive reasoning** – capacity for logical thought, reflection, explanation, justification and communication
- **productive disposition** – habitual inclination to see mathematics as sensible, useful and worthwhile, coupled with a belief in diligence, perseverance and self-efficacy.

Continuity and progression

Junior Cycle

The Junior Cycle Mathematics specification focuses on developing students' mathematical proficiency to enable them to respond to familiar and unfamiliar mathematical situations. The unifying strand in junior cycle draws on mathematical knowledge and skills to solve problems and communicate mathematics.

At senior cycle, students consolidate all prior learning and continue to develop their mathematical proficiency and understanding of the relevance of mathematics in the real world. They build on the rich task-based learning experiences from junior cycle and have opportunities to make connections within mathematics and between mathematics and other disciplines. Students deepen their understanding of mathematical concepts, explore multiple solution strategies, encounter uncertainty, communicate mathematically and develop their ability to generalise and deal with abstraction throughout their two years of senior cycle mathematics.

Beyond senior cycle

Mathematics provides a solid foundation for students to progress to diverse futures, including work, further, adult and higher education, apprenticeships, traineeships and employment, as well as participation in society and adult life. Studying mathematics opens doors to many exciting fields, such as engineering, construction, technology-related jobs, computer science, data science, pharmacy, medicine, education, social sciences, trades, business and finance.

Student learning in senior cycle

Student learning in senior cycle consists of everything students learn **within** all of the subjects and modules they engage with **and** everything students learn which spans and overlaps **across** all of their senior cycle experiences. The overarching goal is for each student to emerge from senior cycle more enriched, more engaged and more competent as a human being than they were when they commenced senior cycle.

For clarity, the learning which spans **across** all of their senior cycle experiences is outlined under the heading 'key competencies'. The learning which occurs **within** a specific subject or module is outlined under the heading 'strands and learning outcomes'. However, it is vital to recognise that key competencies and subject or module learning are developed in an integrated way. By design, key competencies are integrated across the rationale, aims, learning outcomes and assessment sections of specifications. In practice, key competencies are developed by students in schools via the pedagogies teachers use and the environment they develop in their classrooms and within their school. Subjects can help students to develop their key competencies; and key competencies can enhance and enable deeper subject learning. When this integration occurs, students stand to benefit

- during and throughout their senior cycle
- as they transition to diverse futures in further, adult and higher education, apprenticeships, traineeships and employment and
- in their adult lives as they establish and sustain relationships with a wide range of people in their lives and participate meaningfully in society.

When teachers and students make links between the teaching methods students are experiencing, the competencies they are developing and the ways in which these competencies can deepen their subject specific learning, students become more aware of the myriad ways in which their experiences across senior cycle are contributing towards their holistic development as human beings.

Key competencies

Key competencies is an umbrella term which refers to the knowledge, skills, values and dispositions students develop in an integrated way during senior cycle.



Figure 1 The components of key competencies and their desired impact

The knowledge which is specific to this subject is outlined below under 'strands of study and learning outcomes'. The epistemic knowledge which spans across subjects and modules is incorporated into the key competencies.

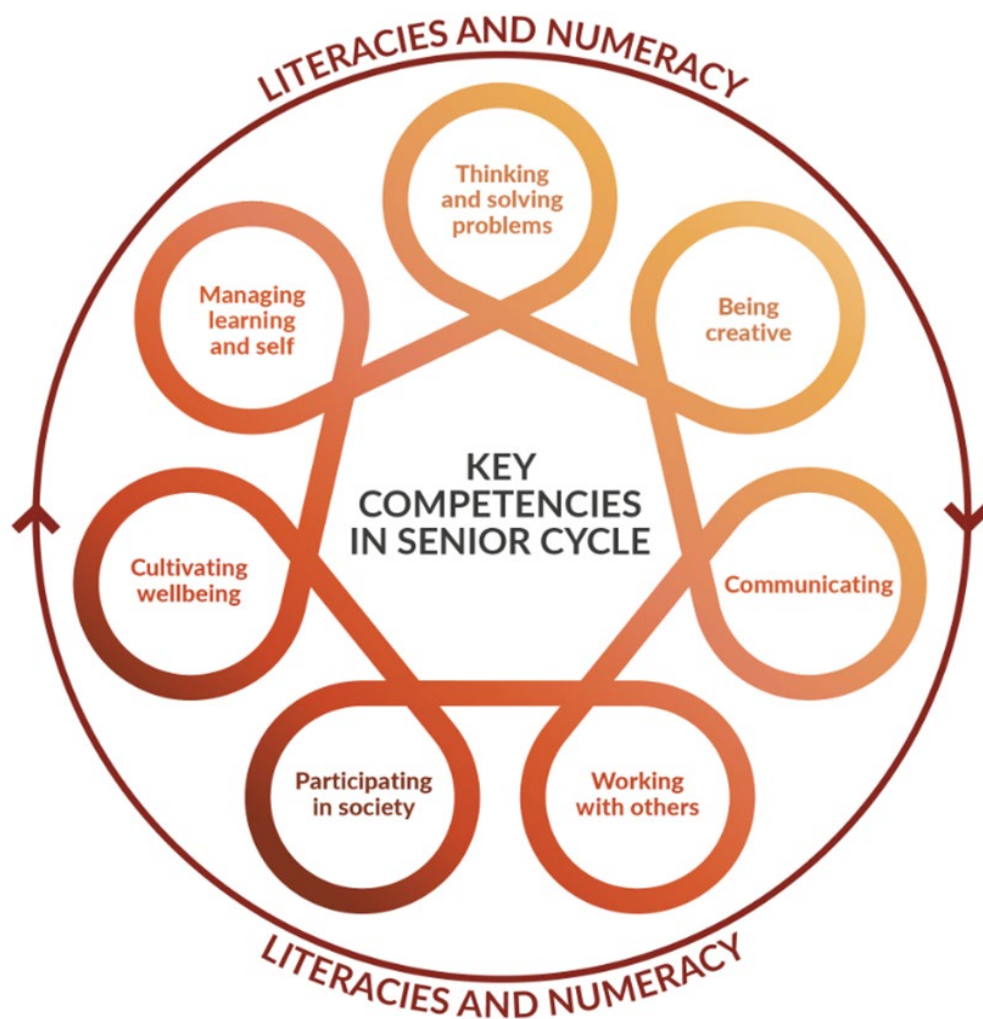


Figure 2 Key Competencies in Senior Cycle, supported by literacies and numeracy

These competencies are linked and can be combined; can improve students' overall learning; can help students and teachers to make meaningful connections between and across different areas of learning; and are important across the curriculum.

The development of students' literacies and numeracy contributes to the development of competencies and vice-versa. Key competencies are supported when students' literacies and numeracy are well developed and they can make good use of various tools, including technologies, to support their learning.

The key competencies come to life through the learning experiences and pedagogies teachers choose and through students' responses to them. Students can and should be helped to develop their key competencies irrespective of their past or present background, circumstances or experiences and should have many opportunities to make their key competencies visible. Further detail in relation to key competencies is available at <https://ncca.ie/en/senior-cycle/senior-cycle-redevelopment/student-key-competencies/>

The key competencies can be developed in Leaving Certificate Mathematics in a range of ways. For example, students develop the key competency of **Being creative** as they explore unfamiliar problems, pose mathematical questions and investigate patterns in numerical, algebraic and geometric contexts. As they attempt to solve problems, such as optimising area, exploring functions or investigating statistical relationships, students are encouraged to try different approaches, represent ideas in multiple ways and adapt strategies when initial methods are unsuccessful. Creativity is further developed as students design and refine mathematical models, test conjectures and translate ideas between representations such as graphs, tables, equations and diagrams.

Students develop the key competency of **Thinking and solving problems** through engagement with mathematical reasoning, modelling and proof. They access, interpret and process information in both familiar and unfamiliar contexts, including financial mathematics, statistics, geometry and functions. Students pose questions, identify relevant information, simplify complex problems and select appropriate strategies to investigate solutions. They use evidence, logical reasoning and mathematical structure to justify conclusions and critically evaluate the validity of results, including consideration of assumptions, limitations and alternative approaches. Through this process, students develop persistence, accuracy and a disciplined approach to reasoning.

Communicating and Working with others is central to learning mathematics. Students develop the key competency of communicating as they explain their reasoning, justify solutions and present mathematical arguments using appropriate notation, terminology and representations. They learn to express ideas clearly in written, verbal and graphical forms and to interpret and critique the reasoning of others. Working with others provides opportunities for students to compare solution strategies, discuss conjectures, refine approaches and build shared understanding of mathematical ideas. Through collaborative problem-solving, students develop an appreciation of different perspectives, learn to negotiate approaches and gain confidence in articulating and defending their reasoning using evidence.

Managing learning and self is developed as students engage with challenging and unfamiliar mathematical tasks that require persistence, reflection and resilience. Students learn to set goals, monitor their progress and respond constructively to feedback. They develop confidence in their ability to improve through effort and reflection and learn that mistakes are a natural part of the problem-solving process. By experiencing success through sustained effort, students develop a sense of achievement and agency in their learning.

Students develop the key competency of **Participating in society** through the use of data, risk, finance and mathematical modelling to evaluate evidence and inform decisions. Students learn to interpret quantitative information, assess claims using evidence and appreciate the role of mathematics in everyday life to support informed engagement with societal issues.

Literacies and numeracy underpin the development of key competencies in Leaving Certificate Mathematics. Literacy involves students interpreting and expressing mathematical ideas using appropriate language, notation and representations across written, verbal, visual and digital forms. Numeracy is developed as students work with quantities, data, models and relationships in contexts such as financial mathematics, statistical analysis and functional relationships. Students interpret data sets, construct representations and critically evaluate mathematical information to support decision-making and reasoning.

Students have multiple opportunities throughout the specification to develop key competencies as they engage actively with the learning outcomes, particularly when supported by the unifying strand, which emphasises problem-solving, reasoning, representation and proof as central mathematical processes.

Strands of study and learning outcomes

This Leaving Certificate Mathematics specification is designed for a minimum of 180 hours of class contact time. It sets out the learning in five strands: a unifying strand—Working Mathematically, and four contextual strands.

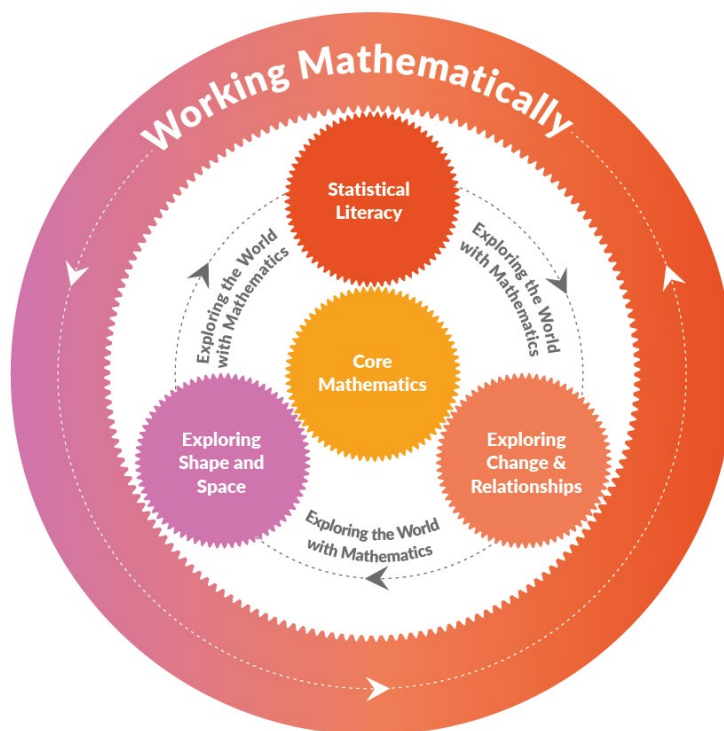


Figure 3 Overview of Leaving Certificate Mathematics

The Working Mathematically strand underpins mathematics and so is considered a unifying strand (Figure 3); it permeates all the strands of the specification. The learning outcomes in this strand identify the knowledge, skills and values related to mathematical practices which are essential to students' learning *about* mathematics throughout the course, underpinning the activities and content in the other strands.

The learning outcomes in the other four contextual strands—(1) Core mathematics: number and algebra (2) Exploring shape and space: geometry and trigonometry, (3) Statistical literacy: investigating with data and (4) Exploring change and relationships: functions and calculus—identify the knowledge *of* mathematics; i.e. the core concepts, principles and procedures that students will learn to apply accurately and logically as they solve problems, explain patterns and make reasoned and justified decisions throughout the course.

An overview of each strand is provided below, followed by a table. The right-hand column contains learning outcomes which describe the knowledge, skills, values and dispositions students should be able to demonstrate after a period of learning. The left-hand column outlines specific areas that students learn about. Taken together, these provide clarity and coherence with the other sections of the specification.

Table 1 Design of learning outcomes for ordinary and higher level

Ordinary level	Higher level
• The learning outcomes present mathematical ideas that are meaningful and relevant and allow for the gradual introduction of abstract ideas, leading students towards the use of mathematics in the context of further study • Students experience mathematics in an exploratory and reflective manner, making connections within mathematics and across other subjects as they apply mathematical ideas to solve some abstract problems and relevant contextualised¹ problems. • Students demonstrate proficiency by choosing appropriate methods, performing calculations precisely and communicating results clearly, demonstrating reliability, attention to detail and the ability to apply mathematics confidently in practical, real-world situations.	• The learning outcomes present mathematical ideas that are meaningful and relevant and allow for the development of powers of abstraction, generalisation and the idea of rigorous proof, providing a solid foundation for further mathematical study. • Students experience mathematics in an exploratory and reflective manner, making connections within mathematics and across other subjects as they apply mathematical ideas fluently to solve both abstract and contextualised problems. • Students demonstrate proficiency by purposely choosing strategies, performing calculations rigorously and communicating results clearly and precisely, demonstrating reliability, attention to detail and the ability to apply mathematics confidently in practical, real-world situations.

¹ Contextualised problems are problems set in real-world or meaningful contexts that can be modelled mathematically and require interpretation of solutions in context.

Learning outcomes have been designed to be achievable relative to students' individual aptitudes and abilities. The outcomes specified at each level, therefore, offer continuity and progression from the corresponding level of mathematics at junior cycle. Learning outcomes promote teaching and learning experiences that develop students' knowledge, skills, values and dispositions incrementally, enabling them to apply their key competencies to different situations as they progress. Students studying at each level will critically engage with mathematics that respects their stage of development. The context, information and results arising from that engagement will build a solid foundation for their post school lives.

Appendix 1 sets out a glossary of action verbs used in the Learning Outcomes.

Unifying Strand: Working mathematically

This strand supports students in developing an appreciation of mathematics as an interconnected body of ideas and reasoning processes, grounded in foundational concepts and skills and characterised by abstraction, structure and rigour. Central to this strand is mathematical problem-solving, understood as an iterative process through which students explore ideas, refine approaches and reflect on the effectiveness of their reasoning.

As they learn to work mathematically, students develop habits of mind that support them in selecting appropriate strategies and critically reviewing their work. They develop an understanding of the role of conjecture, argument and proof in mathematics and build confidence and perseverance as they analyse, interpret, pose and solve problems. Students learn to represent ideas mathematically, translate flexibly between representations, generalise patterns and observations, carry out procedures accurately, and justify and communicate their reasoning. They also learn to evaluate the validity and limitations of their approaches and solutions, including the assumptions underlying mathematical models.

Across the strands, students explore mathematics in relevant contexts, posing questions that can be addressed mathematically. They develop an appreciation of the importance of mathematics in understanding and influencing the world around them, including its role in models that inform decision-making in society and impact people's lives.

Through these experiences, it is hoped that they will perceive mathematics as a rewarding endeavour and develop a sense of achievement through persistence and problem-solving.

Unifying Strand Learning outcomes

Students learn about	Students should be able to
U1. The nature of mathematical knowledge • mathematics as a study of numbers, shapes, measurement and patterns, with mathematical ideas that are connected and build on each other • mathematics as a valuable endeavour underpinned by logic and deductive reasoning	1. appreciate how mathematicians work and how mathematical ideas are cumulative and connected

Students learn about	Students should be able to
U2. The problem-solving process • posing and formulating problems ○ decomposing the problem into manageable parts ○ looking for structure and similarities with other familiar problems ○ researching the background to a problem, where necessary, to analyse factors or variables that affect the situation ○ determining relevant information ○ simplifying the problem using other representations and/or appropriate assumptions • formulating a problem mathematically (where necessary) and/or finding a strategy ○ insightful use of words, diagrams, graphs, tables and equations ○ choosing strategies, comparing strategies individually and collaboratively and considering alternatives • finding solutions and communicating effectively ○ verifying proposed solutions using different strategies and technology ○ communicating mathematical ideas to a range of audiences	1. recognise and pose problems that can be solved with mathematics 2. analyse and simplify a given problem, using appropriate strategies 3. select efficient strategies and justify their use to engage with the problem 4. translate flexibly between different representations of the same idea 5. generalise patterns and observations 6. formulate a situation mathematically 7. use mathematical procedures accurately 8. appropriately scale axes when representing mathematics graphically 9. evaluate mathematical reasoning, results, methods and conclusions 10. communicate mathematical ideas accurately using the language and notation of mathematics

Students learn about	Students should be able to
- evaluating and justifying steps to proposed solutions - using estimation, approximation and verification as problem-solving tools and when evaluating the reasonableness of any proposed solution - determining if a proposed solution is a valid solution and, if not, why not - comparing alternative proposed solutions	11. evaluate proposed solutions to a problem
U3. Generating and using proof - proof as a logical argument establishing that a statement holds in all cases - deductive reasoning - counterexamples and contradiction	- appreciate that proof - is a sequence of logical steps - builds mathematical arguments - is fundamental in mathematics - generate mathematical argument and proofs by - sequencing logical statements - using algebraic manipulation to logically deduce a conclusion - using known results to logically deduce a conclusion - evaluate mathematical arguments to determine whether or not they constitute a proof
U4. Mathematics in society the relevance and importance of mathematics in the world and their everyday lives	- relate mathematics and the real world - apply mathematics in cross-curricular contexts

Strand 1: Core mathematics: number and algebra

This strand identifies number, algebra, pattern and probability as core mathematical concepts that, together with the unifying strand, support progression and coherence across the specification. It emphasises the interconnected nature of these ideas and their role in developing students' mathematical understanding across all strands.

Working from the premise that algebra is an extension and generalisation of arithmetic, emerging naturally from operations on numbers, the strand makes this connection explicit. Building on their junior cycle experience, students extend their understanding of number systems and arithmetic operations, while developing reasoning about mathematical structure, patterns and equivalence. This reasoning provides a foundation for formal algebraic thinking and manipulation. At higher level students are introduced to logarithms as a means of simplifying complex problems.

At senior cycle, the focus shifts towards developing fluency and depth of understanding. Students encounter increasingly complex expressions, equations and relationships and learn to recognise and exploit the structure of mathematical expressions. Transforming expressions into equivalent forms is presented as a powerful problem-solving strategy, supporting the solution of equations, inequalities and systems. Students are introduced to the Factor Theorem as an established method of identifying and justifying roots of polynomial equations, linking algebraic structure with solution strategies and supporting the development of mathematical reasoning and proof.

Students apply their learning in both abstract and real-world contexts, including financial decision-making and problem-solving involving patterns, sequences, counting and probability. Emphasis is placed on selecting appropriate strategies, justifying solutions, interpreting results and communicating mathematical reasoning with precision. Overall, this strand prepares students for further study by developing strong conceptual understanding and reasoning skills.

Strand 1 Learning outcomes

Higher level

Students learn about	Students should be able to
1.1 Core number - fluently and purposefully operating in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ - fractions, percentages, decimals - recurring decimals - multiples, factors - prime numbers Units of measure - metric units of length, area, volume, capacity, mass, - time, - currencies and exchange rates - rates, including speed and unit cost (e.g. km/hr, cm/s, $\text{€}/\text{kg}$) Approximation and accuracy techniques - rounding to a given number of decimal places or significant figures Laws of indices $a^p \cdot a^q = a^{p+q}$ $\frac{a^p}{a^q} = a^{p-q}$ $(a^p)^q = a^{pq}$ $a^0 = 1$ $a^{-p} = \frac{1}{a^p}$ $a^{\frac{1}{q}} = \sqrt[q]{a}$, $q > 0, a > 0$ $a^{\frac{p}{q}} = \sqrt[q]{a^p} = (\sqrt[q]{a})^p$, $q > 0, a > 0$ $(ab)^p = a^p \cdot b^p$ $\left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$ where $p, q \in \mathbb{Z}$, $a, b \in \mathbb{R}$, $a, b \neq 0$ Logarithmic rules $\log_a(xy) = \log_a(x) + \log_a(y)$ $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$ $\log_a(b^x) = x \log_a(b)$	- define and use the number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ appropriately. - translate between - alternate representations of rational numbers - factorised forms - decimal notation and scientific notation ($a \times 10^n$) where $n \in \mathbb{Z}$ and $1 \leq a < 10$ - convert between different units of measure - apply approximation and accuracy techniques when solving problems - apply - the properties of arithmetic operations and relationships between arithmetic operations - units of measure - equivalent representations of numbers - the laws of indices, surds and logarithmic rules - factorials ($n!$) - ratio and proportional reasoning and combinations of these to contextualised and abstract problems, and to make informed financial decisions in everyday contexts - justify proposed solutions to contextualised and abstract problems

Students learn about	Students should be able to
Everyday financial contexts • budgeting, bills, saving, investing, making value for money judgements • tax, PRSI and USC • compound interest, depreciation • interest rates, converting and using monthly/annual interest rates • present value	
1.2 Core algebra Expressions • exploiting structure as an efficient problem-solving strategy • algebraic manipulation as a system of tools for transforming structure while preserving equivalence • transforming algebraic expressions • applying rules and properties ○ arithmetic operations (addition, subtraction, multiplication, division) and their properties (associative and distributive) ○ laws of indices and surds ○ factorisation, including completing the square • structured algebraic reasoning	1. formulate, interpret and evaluate algebraic expressions that represent real-life quantities set in context 2. manipulate the structure of algebraic expressions, including sums, products, factors, quotients and powers, to simplify and factorise expressions 3. justify equivalent forms of algebraic expressions 4. manipulate algebraic expressions • linear expressions in one or more variables with coefficients in $\mathbb{R}$ • quadratic expressions in one or two variable(s) with coefficients in $\mathbb{R}$ • polynomial expressions • rational expressions of the form ○ $\frac{ax+b}{cx+d} \pm \frac{px+q}{rx+s}$, where $a, b, c, d, e, f, p, q, r \in \mathbb{Z}$

Students learn about	Students should be able to
Equations - algebraic equations as mathematical statements of equality between two algebraic expressions - algebraic strategies - completing the square - factorisation - quadratic formula - algebraic strategies - equating coefficients - the factor theorem	- relate algebraic equations to functions - justify proposed solutions to algebraic equations and systems of equations across: - quadratic equations in one or two variable(s), with coefficients in $\mathbb{R}$ - simultaneous linear equations with up to 3 unknowns, with coefficients in $\mathbb{R}$ - one linear equation and one equation of order 2 with two unknowns, with coefficients in $\mathbb{R}$ - cubic equations with at least one integer root, with coefficients in $\mathbb{Z}$ - rational equations of the form $f(x) = g(x)$, with $f(x) = \frac{ax+b}{cx+d} \pm \frac{px+q}{rx+s}, \quad g(x) = \frac{e}{f}$ where $a, b, c, d, e, f, p, q, r, s \in \mathbb{Z}$ - modulus equations of the form $ax + b = c$, where $a, b, c \in \mathbb{R}$ selecting suitable strategies (graphical, numerical and algebraic) - rearrange formulae to highlight a quantity of interest in contextualised and abstract situations
a formula as an algebraic equation solved to show how one quantity depends on others	

Students learn about	Students should be able to
- cross-curricular application of rearranging formulae in contextual situations, including financial and statistical formulae Inequalities - algebraic inequalities as statements of comparison between two expressions	8. interpret and evaluate abstract and contextualised algebraic inequalities 9. justify and represent proposed solution sets, found using suitable graphical, numerical and algebraic strategies, to inequalities of the form: $f(x) < k, f(x) \leq k$ $f(x) > k, f(x) \geq k$ where $f(x)$ is an algebraic expression of the form: $f(x) = ax + b$ $f(x) = \frac{ax+b}{cx+d}$ $f(x) = ax^2 + bx + c$ where $a, b, c, d, k \in \mathbb{Q}$
1.3 Core pattern exploration - patterns in sequences - continuing sequences - generalising	1. investigate patterns in sequences of numbers or objects arising from a process, explaining patterns and relationships in algebraic form 2. describe - arithmetic sequences - quadratic sequences - geometric sequence 3. represent sequences in numerical, visual, tabular, generalised and contextual forms

Students learn about	Students should be able to
- generating terms and defining sequences explicitly Arithmetic sequences $T_n = a + (n - 1)d$, where d is the common difference and a, the initial term Quadratic sequences $T_n = an^2 + bn + c$, where $a, b, c \in \mathbb{Z}$ and $2a$ is the common second difference Geometric sequences $T_n = a \cdot r^{n-1}$, where r is the common ratio and a, the initial term - generating terms and defining sequences recursively Recursive definitions - arithmetic definition: $T_n = T_{n-1} + d$, where d is the common difference and $n \geq 2$ - geometric definition: $T_n = r \cdot T_{n-1}$, where r is the common ratio and $n \geq 2$ - connecting explicit and recursive forms of sequences	- define arithmetic, quadratic and geometric sequences explicitly by finding the n^{th} term - use recursive definitions, for arithmetic and geometric sequences, to model step-by-step change in contextualised situations - convert recursive forms of arithmetic and geometric sequences to explicit forms

Students learn about	Students should be able to
- finding sums of sequences: - arithmetic sequence: $S_n = \frac{n}{2}[2a + (n - 1)d]$ where d is the common difference and a, the first term - geometric sequence: $S_n = \frac{a(1-r^n)}{1-r}$ where r is the common ratio and a is the initial term - limits of sequences $\lim_{n \rightarrow \infty} a_n = L$	- find the sum of the first n terms of arithmetic and geometric sequences - use standard sequence and limit notation - identify whether a sequence has a limit, using appropriate mathematical language and symbols.
1.4 Core probability - the formal language of probability - the concept of equally likely outcomes Combinatorics - counting the arrangements of n distinct objects ($n!$) - permutations: counting the number of ways of arranging r objects from n distinct objects	- apply probability concepts such as set theory, experiments/trials, outcomes, sample spaces to contextualised and abstract problems - use permutations, combinations, set theory and tree diagrams to solve contextualised and abstract problems

Students learn about	Students should be able to
- combinations: counting the number of ways of selecting r objects from n distinct objects Basic rules of probability - general addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ - mutually exclusive events, those that cannot occur at the same time $P(A \cap B) = 0$ - independent events, those where the occurrence of one does not affect the probability of the other event occurring $P(A \cap B) = P(A)P(B)$ - the implications of probability rules for understanding the representativeness of random sampling - conditional probability as the probability that an event A occurs given that B occurs $P(A B) \neq P(B A)$ Expected value - as the average outcome you would expect if an experiment were repeated many times - expected value not necessarily being one of the outcomes	3. explore real-life events using the basic rules of probability 4. apply the basic rules of probability to contextualised and abstract problems involving sampling with or without replacement 5. apply the concept of conditional probability to justify proposed solutions to contextualised and abstract problems $P(A B) = \frac{P(A \cap B)}{P(B)}$ 6. apply the concept of expected value to - establish whether games are fair - quantify risk and support decision making in practical contexts

Students learn about	Students should be able to
Bernoulli trials - an experiment with 2 outcomes classified as 'success' and 'failure' $P(\text{success}) = p$ $P(\text{failure}) = q = 1 - p$ Binomial experiment - fixed number (n) of independent Bernoulli trials - the probability of success in each trial is a constant (p) $P(X = x) = \binom{n}{x} p^x q^{n-x}$ where X: the number of successes in n trials x: a possible value of X The normal distribution - the distinction between discrete and continuous distributions - the standard normal distribution (the bell curve) - the role of mean and standard deviation - the standardisation formula - the interpretation of values within 1, 2 or 3 standard deviations of the mean	7. investigate a sequence of Bernoulli trials and calculate the probability - that the 1st success occurs on the k^{th} trial (geometric) - of k successes in n repeated trials (binomial) - that the k^{th} success occurs on the n^{th} trial (negative binomial) 8. apply the concept of the normal distribution to justify proposed solutions to contextualised problems 9. use probability to evaluate risk in the context of real-life situations

Students learn about	Students should be able to
1.1 Core number - fluently and purposefully operating in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ - fractions, percentages, decimals - recurring decimals - multiples, factors - prime numbers	- define and use the number sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ appropriately. - translate between - alternate representations of rational numbers - factorised forms - decimal notation and scientific notation ($a \times 10^n$) where $n \in \mathbb{Z}$ and $1 \leq a < 10$
Units of measure - metric units of length, area, volume, capacity, mass, - time, - currencies and exchange rates - rates including speed and unit cost (e.g. km/hr, cm/s, $\text{€}/\text{kg}$)	convert between different units of measure
Approximation and accuracy techniques rounding to a given number of decimal places or significant figures	apply approximation and accuracy techniques when solving problems
Laws of indices $a^p \cdot a^q = a^{p+q}$ $\frac{a^p}{a^q} = a^{p-q}$ $(a^p)^q = a^{pq}$ $a^0 = 1$ $a^{-p} = \frac{1}{a^p}$ $a^{\frac{1}{q}} = \sqrt[q]{a}$, $q > 0, a > 0$ $(ab)^p = a^p \cdot b^p$ $\left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$ where $p, q \in \mathbb{Z}$, $a, b \in \mathbb{R}$, $a, b \neq 0$	- apply - the properties of arithmetic operations and relationships between arithmetic operations - units of measure - equivalent representations of numbers - the laws of indices, surds and factorials ($n!$) - ratio and proportional reasoning and combinations of these to contextualised and abstract problems, and to make informed financial decisions in everyday contexts
Everyday financial contexts - budgeting, bills, saving, investing, making value for money judgements - tax, PRSI and USC - compound interest, depreciation - interest rates, using monthly/annual interest rates	justify proposed solutions to contextualised and abstract problems

Students learn about	Students should be able to
1.2 Core algebra Expressions - exploiting structure as an efficient problem-solving strategy - algebraic manipulation as a system of tools for transforming structure while preserving equivalence - transforming algebraic expressions - applying rules and properties - arithmetic operations (addition, subtraction, multiplication, division) and their properties (associative and distributive) - laws of indices and surds - factorisation - structured algebraic reasoning Equations - algebraic equations as mathematical statements of equality between two algebraic expressions	- formulate, interpret and evaluate algebraic expressions that represent real-life quantities set in context - manipulate the structure of algebraic expressions, including sums, products, factors, quotients and powers, to simplify and factorise expressions - identify equivalent forms of algebraic expressions - manipulate algebraic expressions - linear expressions in one or more variables with coefficients in $\mathbb{R}$ - quadratic expressions in one variable with coefficients in $\mathbb{Z}$ - rational expressions of the form $\frac{a}{bx+c} \pm \frac{p}{qx+r},$ where $a, b, c, d, p, q, r \in \mathbb{Z}$ - relate algebraic equations to functions

Students learn about	Students should be able to
- algebraic strategies - factorisation - quadratic formula - form equations given whole number roots - cross-curricular applications of rearranging formula in contextual situations, including financial and statistical formulae Inequalities - algebraic inequalities as statements of comparison between two expressions	- justify proposed solutions to algebraic equations and systems of equations, with coefficients in $\mathbb{Z}$, across: - quadratic equations in one variable - simultaneous linear equations with 2 unknowns - one linear equation and one equation of order 2 with two unknowns (where either the coefficient of x or y in the linear equation is ± 1) - rational equations of the form $f(x) = g(x)$, with $f(x) = \frac{a}{bx + c} \pm \frac{p}{qx + r}, g(x) = \frac{e}{f},$ where $a, b, c, e, f, p, q, r \in \mathbb{Z}$ selecting suitable strategies (graphical, numerical and algebraic) - rearrange formulae to highlight a quantity of interest in contextualised and abstract situations - interpret and evaluate abstract and contextualised algebraic inequalities - justify and represent proposed solution sets, found using suitable graphical, numerical and algebraic strategies, to inequalities of the form: $f(x) < k, f(x) \leq k$ $f(x) > k, f(x) \geq k$

Students learn about	Students should be able to
	where $f(x)$ is an algebraic expression of the form: $f(x) = ax + b$, where $a, b, k \in \mathbb{Q}$
1.3 Core pattern exploration - patterns in sequences - continuing sequences - generalising - generating terms and defining sequences explicitly Arithmetic sequences $T_n = a + (n - 1)d$, where d is the common difference and a, the initial term Quadratic sequences $T_n = an^2 + bn + c$, where $a, b, c \in \mathbb{Z}$ and $2a$ is the common second difference Geometric sequences $T_n = a \cdot r^{n-1}$, where r is the common ratio and a, the initial term	- investigate patterns in sequences of numbers or objects arising from a process, explaining patterns and relationships in algebraic form - recognise and describe - arithmetic sequences - quadratic sequences - geometric sequences - represent sequences in numerical, visual, tabular, generalised and contextual forms - define arithmetic, quadratic and geometric sequences explicitly by finding the n^{th} term

Students learn about	Students should be able to
- generating terms and defining sequences recursively Recursive definitions - arithmetic definition: $T_n = T_{n-1} + d$ where d is the common difference and $n \geq 2$ - geometric definition: $T_n = r.T_{n-1}$, where r is the common ratio and $n \geq 2$ - connecting explicit and recursive forms of sequences	- use recursive definitions for arithmetic and geometric sequences to model step-by-step change in contextual situations - convert recursive forms of arithmetic and geometric sequences to explicit forms
1.4 Core probability Combinatorics - counting the arrangements of n distinct objects ($n!$) - permutations: counting the number of ways of arranging r objects from n distinct objects Basic rules of probability - general addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ - mutually exclusive events, those that cannot occur at the same time $P(A \cap B) = 0$	- apply probability concepts such as set theory, experiments/trials, outcomes, sample spaces to contextualised and abstract problems - use permutations, combinations, set theory and tree diagrams to solve contextualised and abstract problems - explore real-life events using the basic rules of probability

Students learn about	Students should be able to
- independent events, those where the occurrence of one does not affect the probability of the other event occurring $P(A \cap B) = P(A)P(B)$ - the implications of probability rules for understanding the representativeness of random sampling Bernoulli trial as - an experiment with 2 outcomes classified as 'success' and 'failure' $P(\text{success}) = p$ $P(\text{failure}) = q = 1 - p$ The normal distribution - the distinction between discrete and continuous distributions - the standard normal distribution (the bell curve) - the role of mean and standard deviation - the standardisation formula - the interpretation of values within 1, 2 or 3 standard deviations of the mean	- investigate the probability of two independent events occurring - apply the basic rules of probability to contextualised problems involving sampling with or without replacement - investigate a sequence of up to 3 Bernoulli trials and calculate the probability - that the 1st success occurs on the k^{th} trial (geometric) - of k successes in n repeated trials (binomial) - apply the concept of the normal distribution to justify proposed solutions to contextualised problems - use probability to evaluate risk in the context of real-life situations

Strand 2: Exploring shape and space: geometry and trigonometry

In this strand, students use tools such as dynamic geometry software and the coordinate plane to transform abstract ideas about shape and space into concrete, visual and measurable concepts. This provides a unified framework for connecting geometric understanding with algebraic reasoning.

Exploring 2D shapes and 3D solids in their environment provides opportunities to develop spatial reasoning and the analytical skills required to model and solve problems in science, technology, cryptography, architecture and navigation.

Building on foundational knowledge, students consolidate their understanding of Euclidean geometry encountered in junior cycle and deepen their insights into how algebra and geometry work together. They engage in geometric reasoning by applying established theorems and using these results to justify solutions to problems.

Within the coordinate plane, students learn how algebraic equations describe geometric shapes and explore geometric properties to reinforce algebraic relationships, emphasising mathematical processes and logical reasoning over isolated calculations.

The algebraic-geometric connection is further explored through trigonometry as students analyse and solve problems involving angles, triangles and periodic phenomena (HL only). They experience trigonometric functions as powerful tools for modelling phenomena that behave repetitively, rhythmically or in circular motion. On the coordinate plane, these relationships become visible through graphing, while algebraic manipulation reveals their underlying patterns and applications in the real world. By applying 3D trigonometry to real-world situations, students working at higher level see the interconnected nature of mathematics, using ideas from different areas of mathematics to generate and solve complex, practical problems.

Throughout the strand, students engage meaningfully with the problem-solving cycle, learning to formulate conjectures, test hypotheses and communicate their reasoning through multiple, interconnected representations.

Students learn about	Students should be able to
2.1 Deductive reasoning in geometry - the concept of mathematical proof - use of symbolic notation and precise mathematical language - application of axioms 1-6, corollaries 1-5 and theorems, 1-20, from <i>Geometry for Post-primary School Mathematics</i>.	apply step-by-step deductive reasoning based on geometric results to justify proposed solutions in both abstract and contextualised problems
2.2 Geometry within the coordinate plane - using appropriate graphing technology Geometric shapes - lines: $y = mx + c$ slope m, intercept c for $m, c \in \mathbb{R}$ $ax + by + c = 0$ for $a, b, c \in \mathbb{R}$ - circles: centre (h, k), radius r $(x - h)^2 + (y - k)^2 = r^2$ for $h, k, r \in \mathbb{R}, r > 0$ Geometric concepts - distance between points - perpendicular distance between a point and a line - midpoint of a line segment - slope/gradient of a line - slopes of parallel and perpendicular lines - point of intersection of lines and y-intercept of lines - centre and radius of circles	- relate mathematical equations to geometric shapes formed by lines or circles - sketch and draw lines and circles given algebraic equations - generate algebraic equations from representations of lines and circles in the coordinate plane - translate between different representations (algebraic, graphical, contextual) of lines, circles and sections of these - apply - algebraic reasoning - appropriate formulae - translations and reflections to problems in the coordinate plane involving lines and circles, within the scope of the geometric concepts and properties listed in this section

Students learn about	Students should be able to
- tangents of circles - collinear points - translations of points and shapes - reflections and line symmetry Geometric properties - sum of angles in a triangle - areas and perimeters - properties of quadrilaterals - congruence and similarity	justify proposed solutions to contextualised and abstract problems involving lines and circles, using the geometric concepts and properties in this section
2.3 Trigonometry - three figure bearings as angles measured clockwise from North Formulae - sin, cos and tan as ratios of side lengths in right-angled triangles - the inverse functions arcsin, arccos and arctan $\tan \theta = \frac{\sin \theta}{\cos \theta}$ - Pythagoras' Theorem - sine rule, cosine rule - area of a triangle $= \frac{1}{2} ab \sin C$ - angles of depression/elevation - arc length and area of sectors - using 3D physical models and dynamic geometry software as visualisation tools - generating their own 3D problems - the definition of sin and cos as functions of a real variable that track coordinates of points on the unit circle - functions of the form f where: $f(t) = A \sin(\omega t + \alpha) \pm c$ $f(t) = A \cos(\omega t + \alpha) \pm c$ for all $t \in \mathbb{R}$, where A, ω, α, c are constants in $\mathbb{R}$	- apply three figure bearings and appropriate formulae to solve abstract and contextualised problems involving heights, distances, angles and areas - justify proposed solutions to problems involving heights, distances, angles and areas using appropriate geometrical reasoning - convert between degree measure and radian measure of angles and use both appropriately in context - investigate contextualised problems involving the application of 3D trigonometry, justifying strategies using appropriate mathematical reasoning - relate trigonometric equations to functions

Students learn about	Students should be able to
Real-world applications of trigonometric functions - parameters: amplitude, period, frequency, vertical and horizontal shift - real-life contexts /cross-curricular links such as - swinging pendulum - Ferris wheel - mass on a spring - tides - sound waves - alternating current	- model real-world situations using trigonometric functions - relate the parameters A, ω, α and c of the above functions, and the impact of changing them, to the real-life context

Students learn about	Students should be able to
2.1 2D Shapes and 3D solids • nets; 2D representations of 3D solids • using physical models, physical representations and dynamic geometry software to investigate the relationship between 2D shapes and 3D solids Formulae • perimeter, area and circumference of 2D shapes ◦ parallelogram, triangle, trapezium, circle • surface area, volume and capacity of 3D solids ◦ cuboid, cube, cylinder, sphere and hemisphere, cone, triangular prism and other shapes with uniform cross section	1. relate 2D shapes and 3D solids using nets 2. apply appropriate formulae to solve abstract and contextualised problems involving 2D shapes and 3D solids, and justify proposed solutions using appropriate mathematical reasoning 3. investigate contextualised problems involving the decomposition and re-composition of shapes and solids, and justify strategies using clear mathematical reasoning
2.2 Deductive reasoning in geometry • the concept of mathematical proof • use of symbolic notation and precise mathematical language • application of axioms 1-6, corollaries 1-5 and theorems, 1-20, from <i>Geometry for Post-primary School Mathematics</i>.	1. apply step-by-step deductive reasoning based on geometric results to justify proposed solutions in both abstract and contextualised problems

Students learn about	Students should be able to
2.3 Geometry within the coordinate plane - using appropriate graphing technology Geometric shapes - lines: $y = mx + c$ slope m, intercept c for $m, c \in \mathbb{R}$ $ax + by + c = 0$ for $a, b, c \in \mathbb{R}$ - circles: centre $(0,0)$, radius r $x^2 + y^2 = r^2$ for $r \in \mathbb{R}, r > 0$ Geometric concepts - distance between points - perpendicular distance between a point and a line - midpoint of a line segment - slope/gradient of a line - slopes of parallel and perpendicular lines - point of intersection of lines and y-intercept of lines - centre and radius of circles - tangents of circles - collinear points - translations of points and shapes - reflections and line symmetry Geometric properties - sum of angles in a triangle - areas and perimeters - properties of quadrilaterals - congruence and similarity	- relate mathematical equations to geometric shapes formed by lines or circles - sketch and draw lines and circles given algebraic equations - generate algebraic equations from representations of lines and circles in the coordinate plane - translate between different representations (algebraic, graphical, contextual) of lines, circles and sections of these - apply - algebraic reasoning - appropriate formulae - translations and reflections to problems in the coordinate plane involving lines and circles, within the scope of the geometric concepts and properties listed in this section - justify proposed solutions to contextualised and abstract problems involving lines and circles, using the geometric concepts and properties in this section
2.4 Trigonometry three figure bearings as angles measured clockwise from North	apply three figure bearings and appropriate formulae to solve abstract and contextualised problems involving heights, distances, angles and areas

Students learn about	Students should be able to
Formulae • trigonometric ratios for sin, cos and tan as ratios in right-angled • the inverse functions arcsin, arccos and arctan • $\tan \theta = \frac{\sin \theta}{\cos \theta}$ • Pythagoras' Theorem • sine rule, cosine rule • area of a triangle = $\frac{1}{2} ab \sin C$ • angles of depression/elevation	2. justify proposed solutions to problems involving heights, distances, angles and areas using appropriate geometrical reasoning

Strand 3: Statistical literacy: investigating with data

This strand aims to develop statistical literacy, an essential skill in today's data-driven world. Within this specification, statistical literacy is defined as the ability to understand, interpret and critically evaluate data. Students build this literacy by posing genuine questions and exploring real data sets to investigate them.

In practice, statisticians collaborate with researchers to design studies that collect unbiased, representative data to answer research questions objectively. Accordingly, this strand is designed to mirror that process. To support students' engagement, it begins with the exploration of how statistics are used in the media. Students then investigate and apply key concepts through the statistical enquiry cycle, communicating findings from large, multivariate data sets to explore real-world issues and support evidence-based conclusions.

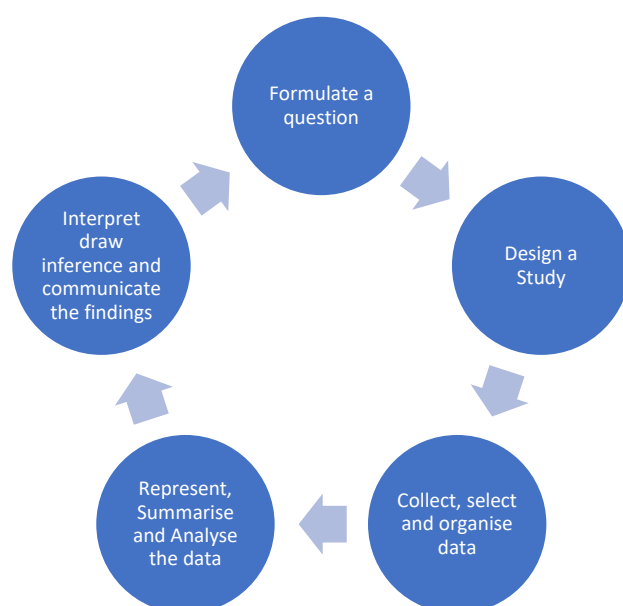


Figure 4 The statistical enquiry cycle

Progressing from the basic statistics experienced in junior cycle to statistical inference is a key milestone in becoming statistically literate. In senior cycle, students are expected to move beyond describing data to understanding the significance of the data collection process and to drawing conclusions and making justified predictions about larger populations.

Distributions play a central role in supporting this learning. Through exploration, students gain insight into the importance of the normal distribution in enabling probability-based reasoning, because of its role in the Central Limit Theorem, which justifies generalising from a

sample to a population. Probability, introduced and developed in Strand 1, underpins this work by providing the theoretical basis for modelling uncertainty in statistics.

As they progress through this strand, students will use real-world multivariate data sets as a basis for exploration. Their learning will be assessed through an Additional Assessment Component described on page 61 of this specification.

Strand 3 Learning outcomes

Higher level

Students learn about	Students should be able to
3.1 Motivation for statistical research studies Sources of statistical claims - newspaper articles, social media, television/radio, scientific journals, research reports - the usefulness of statistics - common ways statistics may be misused or misrepresented - how graphs can be used to mislead	- examine how statistical claims are used in research studies - evaluate the consequences of reporting misleading and/or biased conclusions from research studies
3.2 Principles of study design Fundamental concepts of study design - bias (deliberate, unintentional) representativeness, reliability, validity, confounding variables (correlation vs. causation), control group vs. treatment group Features of a good research hypothesis - unbiased - clear - specific - context-driven and purposeful - can be objectively tested	- recognise the qualities of a good study - classify examples of studies as - sample surveys - questionnaire design - sampling design - designed experiments - observational studies - explain the merits of each of the study types with reference to bias, representativeness, reliability, validity and confounding variables - generate testable hypotheses that - are meaningful, unbiased and relevant - can be objectively tested by a study

Students learn about	Students should be able to
3.3 Sample surveys 3.3 (a) Formulating statistical questions (questionnaire design) - questionnaire bias as systematic errors or distortions that influence measurement and reporting - deliberate bias - unintentional bias - unnecessary complexity - desire to please	- identify the presence and type of bias in a question - construct unbiased survey questions
3.3 (b) Selecting and organising data (sampling design) Variable types - categorical (nominal, ordinal), numerical (discrete, continuous) - census vs. sample survey - sampling frames as the key to unbiased sampling Variability - variability in populations - variability within samples - variation between samples - how sample surveys are designed to understand and explain variability in a population The difference between randomness and representativeness: - random samples are often used in practice, but they are not always the same as representative samples Simple sampling designs within the context of real-life data sets - Simple Random Sampling (SRS) - systematic sampling - stratified sample	- consider the variable types and select appropriate variables relevant to a research question - distinguish between a census and a sample survey - identify an appropriate sampling frame - evaluate survey designs with respect to - representativeness and bias - variability - explore how the accuracy of statistical estimates depends on the variability of the population and the sample size - evaluate sampling mechanisms with respect to - the representativeness of random sampling - when a source may lack detail or show bias - the quality and representativeness of the sample data

Students learn about	Students should be able to
selection bias as systematic errors or distortions that influence sample selection (e.g. unavailability of a sample frame, sampling from the wrong population, the use of volunteers in samples)	use SRS, systematic and stratified sampling techniques to select representative, unbiased data
3.4 Representing and summarising data graphically and numerically Distribution shapes - symmetric, uniform, skewed right, skewed left and normal Measures of central tendency - mean, mode, median Measures of variability - range, IQR, standard deviation Tools for analysing data - line plots, histograms, stem and leaf plots, boxplots Tools for identifying and justifying trends - scatterplots, correlation coefficient	- investigate distributions examining: - the shape of each distribution - how their shape is influenced by centre, spread and skewness - determine appropriate measures of central tendency and variability for use with different shapes of distribution - use appropriate statistical tools to: - describe central tendency and variability - analyse patterns in data and relationships between variables - compare datasets - identify and justify trends, patterns, outliers and relationships
3.5 Statistical inference: analysing, interpreting and drawing inferences from data The foundations of statistical inference using a sample to make inferences about a population	investigate how sampling variability affects the reliability of using sample statistics to make inferences about a population

Students learn about	Students should be able to
Between sample variability - different samples selected from the same population can vary; consequently, sample statistics also vary from sample to sample - the principle of the central limit theorem (the distribution of the sample means approximates a normal distribution) - empirical rule as an approximation of probabilities for the normal distribution	
Using sample proportions to estimate population proportions Measures of accuracy of sample estimates - margin of error² $\frac{1}{\sqrt{n}}$ - how 95% confidence intervals use the critical value $z = 1.96$	- compute the margin of error for various sample sizes, n, where $n \geq 30$ - construct a 95% confidence interval for a population mean and a population proportion
Using sample means to estimate population means how 95% confidence intervals use the critical value $z = 1.96$	
The principles of hypothesis testing - null and alternate hypotheses - a priori hypotheses - significance levels - overlapping margin of error intervals or confidence intervals as a simplified form of hypothesis testing for population proportions - overlapping confidence intervals as a simplified form of hypothesis testing for population means	- appreciate that a hypothesis test uses sample statistics to make decisions about population parameters - perform a hypothesis test to test a claim about the population appropriately using margin of error intervals and confidence intervals - communicate justified conclusions

² The margin of error referred to here is the maximum value of the radius of the 95% confidence interval.

Students learn about	Students should be able to
3.6 Conducting statistical investigations - the statistical enquiry cycle - the role of iteration in the enquiry cycle	apply the full statistical enquiry cycle as a structured process for answering statistical questions

Students learn about	Students should be able to
3.1 Motivation for statistical research studies Sources of statistical claims from research studies • newspaper articles, social media, television/radio, scientific journals, research reports • the usefulness of statistics • common ways statistics may be misused or misrepresented • how graphs can be used to mislead	1. examine how statistical claims are used in research studies 2. evaluate the consequences of reporting misleading and/or biased conclusions from research studies
3.2 Principles of study design Fundamental concepts of study design • bias (deliberate, unintentional) representativeness, reliability, validity, confounding variables (correlation vs. causation), control group vs. treatment group Features of a good research hypothesis • unbiased • clear • specific • context-driven and purposeful • can be objectively tested	1. recognise the qualities of a good study 2. classify examples of studies as • sample surveys ○ questionnaire design ○ sampling design • designed experiments • observational studies 3. explain the merits of each of the study types with reference to bias, representativeness, reliability, validity and confounding variables 4. generate testable hypotheses that • are meaningful, unbiased and relevant • can be objectively tested by a study

Students learn about	Students should be able to
3.3 Sample surveys 3.3 (a) Formulating statistical questions (questionnaire design) - questionnaire bias as systematic errors or distortions that influence measurement and reporting - deliberate bias - unintentional bias - unnecessary complexity - desire to please	- identify the presence and type of bias in a question - construct unbiased survey questions
3.3 (b) Selecting and organising data (sampling design) Variable types - categorical (nominal, ordinal), numerical (discrete, continuous) - census vs. sample survey - sampling frames as the key to unbiased sampling Variability - variability in populations - variability within samples - variation between samples - how sample surveys are designed to understand and explain variability in a population The difference between randomness and representativeness - random samples are often used in practice, but they are not always the same as representative samples Simple sampling designs within the context of real-life data sets - Simple Random Sampling (SRS) - systematic sampling	- consider the variable types and select appropriate variables relevant to a research question - distinguish between a census and a sample survey - identify an appropriate sampling frame - evaluate survey designs with respect to - representativeness and bias - variability - explore how the accuracy of statistical estimates depends on the variability of the population and the sample size - evaluate sampling mechanisms with respect to - the representativeness of random sampling - when a source may lack detail or show bias - the quality and representativeness of the sample data

Students learn about	Students should be able to
• stratified sample • selection bias as systematic errors or distortions that influence sample selection (e.g. unavailability of a sample frame, sampling from the wrong population, the use of volunteers in samples)	7. use SRS, systematic and stratified sampling techniques to select representative, unbiased data
3.4 Representing and summarising data graphically and numerically Distribution shapes • symmetric, uniform, skewed right, skewed left and normal Measures of central tendency • mean, mode, median Measures of variability • range, IQR, standard deviation Tools for analysing data • line plots, histograms, stem and leaf plots, boxplots Tools for identifying and justifying trends • scatterplots, correlation coefficient	1. investigations distributions examining: • the shape of each distribution • how their shape is influenced by centre, spread and skewness 2. determine appropriate measures of central tendency and variability for use with different shapes of distribution 3. use appropriate statistical tools to: • describe central tendency and variability • analyse patterns in data and relationships between variables • compare datasets • identify and justify trends, patterns, outliers and relationships
3.5 Statistical inference: analysing, interpreting and drawing inferences from data The foundations of statistical inference • using a sample to make inferences about a population	1. investigate how sampling variability affects the reliability of using sample statistics to make inferences about a population

Students learn about	Students should be able to
Between sample variability - different samples selected from the same population can vary; consequently, sample statistics also vary from sample to sample - the principle of the central limit theorem (the distribution of the sample means approximates a normal distribution) - empirical rule as an approximation of probabilities for the normal distribution	
Using sample proportions to estimate population proportions measures of accuracy of sample estimates: margin of error ³ $\frac{1}{\sqrt{n}}$	2. compute the margin of error for various sample sizes, n , where $n \geq 30$
The principles of hypothesis testing - null and alternate hypotheses - a priori hypotheses - significance levels - overlapping margin of error intervals as a simplified form of hypothesis testing for population proportions	3. appreciate that a hypothesis test uses sample statistics to make decisions about population parameters 4. perform a hypothesis test to test a claim about the population appropriately using margin of error intervals 5. communicate justified conclusions

³ The margin of error referred to here is the maximum value of the radius of the 95% confidence interval.

Students learn about	Students should be able to
3.6 Conducting statistical investigations - the statistical enquiry cycle - the role of iteration in the enquiry cycle	apply the full statistical enquiry cycle as a structured process for answering statistical questions

Strand 4: Exploring change and relationships: functions and calculus

In this strand, students experience functions as a unifying concept in mathematics, connecting algebraic manipulation, graphical interpretation and real-world applications. Building from junior cycle, students progress from viewing functions as rules that describe relationships between inputs and outputs to understanding functions as mathematical objects in their own right, with an emphasis on notation, structure and properties.

Students develop fluency in representing, analysing and solving problems involving linear, polynomial, exponential, trigonometric and rational functions. They interpret how functions behave and use graphical, algebraic and contextual approaches to model dynamic phenomena.

The strand is designed to support progression from concrete function concepts to more sophisticated analytical techniques, such as function composition and inverse functions. Supported by graphing technology, students move from basic function evaluation to constructing mathematical models that describe real-world phenomena involving rates of change.

At higher level, students develop their understanding of logarithms as an essential tool for analysing exponential growth and decay, enabling them to solve equations, interpret real-world models and understand inverse relationships.

The introduction of differential calculus and integral calculus at higher level marks a significant step in mathematical development. Students learn to study rates of change using differentiation and accumulation using integration, applying techniques such as the power, product, quotient and chain rules for differentiation, as well as integration by substitution. They explore optimisation problems and use integration to calculate areas and solve problems involving displacement, velocity and acceleration, gaining insights into how functions behave both locally and globally.

Students learn about	Students should be able to
4.1 Functions, definitions and representations Function concepts - definition: given two sets A and B, a function from A to B associates to each element of A a single element of B - domain: set of inputs for which the function is defined - codomain: specified set to which the outputs belong - range: set of outputs obtained Function types - exponential functions of the form f given by $f(x) = a^x$ $f(x) = a(e^{bx})$ for all $x \in \mathbb{R}$, where a and b are constants in $\mathbb{R}$ - trigonometric functions of the form f given by $f(x) = a\sin(bx + k) + c$ $f(x) = a\cos(bx + k) + c$ for all $x \in \mathbb{R}$, where a, b, c and k are constants in $\mathbb{R}$ - rational functions of the form f given by $f(x) = \frac{k}{ax + b}$ for all $x \in \mathbb{R}$, where a and b are constants in $\mathbb{R}$ - using graphing technology	- demonstrate an understanding of the concept of a function - identify the domain, codomain and range of a function using appropriate notation - recognise and use correct function notation - identify whether a function is injective, surjective, bijective or none of these - graph and evaluate functions: - polynomial - exponential - trigonometric - rational - relate algebraic, graphical and contextual representations of the above functions

Students learn about	Students should be able to
Function characteristics x and y-intercepts - real roots - slopes of secant lines of the graphs of the function - local maximum and minimum - absolute maximum and minimum - limits and limiting behaviour $\lim_{x \rightarrow \infty} f(x), \lim_{x \rightarrow a} f(x)$ - concept of asymptotes Function transformations - translations - reflections - scalings and combinations of these Functions described graphically without formulae - continuous functions, piecewise functions and those with jumps and discontinuities - choosing suitable strategies, both graphical and algebraic Equations and inequalities of the form $f(x) = k$ $f(x) \leq k$ $f(x) = g(x)$ $f(x) \leq g(x)$ where f and g may be polynomial, exponential, trigonometric, or rational, or where graphs of f and g are given	7. identify and interpret characteristics of functions from their graphs 8. apply function transformations and relate the change in parameters to the graphical transformation 9. relate graphical representations of functions, without formulae, to their real-world contexts 10. determine analytical forms of functions from their graphs 11. justify proposed solutions to contextualised and abstract problems modelled by functions
4.2 Functions and their inverses Function composition the map $f \circ g: x \mapsto f(g(x))$ as a composition function, where the range of g is in the domain of f	1. recognise, form and decompose composite functions 2. determine the largest domain on which a composite function can be defined

Students learn about	Students should be able to
Inverse function concept - the inverse of a function f is a function f^{-1}, where $f^{-1} \circ f(x) = x$, for all x in the domain of f, and $f \circ f^{-1}(x) = x$ for all x in the domain of f^{-1} - the horizontal line test and its limitations - real-world contexts where reversing a process depends on having the correct information (encryption) - the log function as the inverse of the exponential function - functions of the form f given by $f(x) = a(e^{bx})$, for all $x \in \mathbb{R}$ where a, b, are constants in $\mathbb{R}$ Cross-curricular applications of logs - compound interest - pH scale - decibel - Richter scale - growth and decay	- relate a function and its inverse (where an inverse exists) - use suitable strategies (algebraic, graphical) to determine the inverse of a function (where it exists) - use correct notation for inverse functions in context - relate logarithms to exponential functions - justify proposed solutions to contextualised and abstract problems modelled by logarithmic functions
4.3 Differential calculus Exploratory approach to a derivative - using appropriate graphing technologies to explore the average and instantaneous rate of change of a function f - the core conceptual distinction between the average and instantaneous rate of change of a function	- relate the average rate of change of a function over an interval to the slope of a secant line in real-world contexts - interpret the derivative of a function as both the instantaneous rate of change and the slope of the tangent line to a graph

Students learn about	Students should be able to
- a graphical approach to the derivative - the derivative as: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ Derivative notation - first derivative: $\frac{dy}{dx}$, $f'(x)$ - second derivative: $\frac{d^2y}{dx^2}$, $f''(x)$ Differentiation techniques - established derivatives published in the formulae and tables booklet - product rule - quotient rule - chain rule Function behaviour - increasing/decreasing - intervals where increasing/decreasing - local maxima and minima - absolute maximum and minimum values - points of inflection Applications of differentiation - velocity (v) and acceleration (a) as rates of change: $v = \frac{ds}{dt}$ $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$ - optimising contexts: - maximising profit - minimising cost, loss	- use first principles to determine derivatives of linear and quadratic functions - apply appropriate derivative notation - use the table of derivatives to find the derivative of the following functions - polynomial - exponential - trigonometric - rational powers - logarithms - differentiate sums, differences products, quotients and compositions of functions of the above form - relate differentiation to the behaviour of the graph of a function - use differentiation to sketch the graph of a function - use differentiation to justify and interpret proposed solutions to contextualised problems involving optimisation, related rates of change, equations of tangent lines, velocity and acceleration

Students learn about	Students should be able to
○ maximising/minimising area of 2D shapes (circle, quadrilateral) subject to constraints ○ maximising/minimising surface area and volume of 3D solids (cuboid, cube, cylinder, triangular prism, sphere, hemisphere, cone) subject to constraints	
4.4 Integral calculus Integration concepts • integration as the reverse process of differentiation • integration interpreted as area: Riemann sums and accumulated net area, definite and indefinite integrals Integration techniques • evaluating integrals • substitution as an integration technique Applications of integration • average value of a function over an interval • areas of plane regions bounded by polynomial, exponential and trigonometric curves • volumes of solids formed by rotating linear or polynomial functions about the x-axis • displacement, velocity and acceleration, total distance travelled • accumulated change	1. relate the indefinite integral to the family of anti-derivatives of a function and the definite integral $\int_a^b f(x) dx$ to the accumulated net area under the curve $y = f(x)$ between the vertical lines $x = a$ and $x = b$ 2. evaluate integrals of sums, differences and constant multiples of functions of the form f given by • $f(x) = x^a$ • $f(x) = a^x$ • $f(x) = \sin ax$ • $f(x) = \cos ax$ 3. use integration by substitution to evaluate integrals of composite functions of the form $f(ax + b)$ 4. apply integration to justify and interpret proposed solutions to contextualised and abstract problems

Students learn about	Students should be able to
4.1 Functions, definitions and representations Function concepts - definition: given two sets A and B, a function from A to B associates to each element of A a single element of B - domain: set of inputs for which the function is defined - codomain: specified set to which the outputs belong - range: set of outputs obtained Function types - exponential functions of the form f given by $f(x) = a^x$ for all $x \in \mathbb{R}$, where a and b are constants in $\mathbb{R}$ - using graphing technology Function characteristics x and y-intercepts - real roots - slopes of secant lines of the graph of the function - local maximum and minimum - absolute maximum and minimum Function transformations - translations - reflections - stretches and combinations of these - choosing suitable strategies both graphical and algebraic	- demonstrate an understanding of the concept of a function - identify the domain, codomain and range of a function using appropriate notation - use correct function notation - graph and evaluate linear, quadratic, cubic and exponential functions - relate algebraic, graphical and contextual representations of the above functions - identify and interpret characteristics of functions from their graphs - apply function transformations and relate the change in parameters to the graphical transformation - relate graphical representations of functions, without formulae, to their real-world contexts

Students learn about	Students should be able to
Equations and inequalities of the form $f(x) = k$ $f(x) \leq k$ $f(x) = g(x)$ $f(x) \leq g(x)$ where f and g may be linear, quadratic, cubic or exponential, or where graphs of f and g are given	9. justify proposed solutions to contextualised and abstract problems modelled by functions
4.2 Functions and their inverses Inverse function concept the inverse of a function f is a function f^{-1}, where $f^{-1} \circ f(x) = x$, for all x in the domain of f, and $f \circ f^{-1}(x) = x$ for all x in the domain of f^{-1}	1. relate a function and its inverse (where an inverse exists)
4.3 Differential calculus Exploratory approach to a derivative - using appropriate graphing technologies to explore the average and instantaneous rate of change of a function $f(x)$ - the core conceptual distinction between the average and instantaneous rate of change of a function - a graphical approach to the derivative Derivative notation - first derivative: $\frac{dy}{dx}, f'(x)$ Differentiation techniques - power rule: $\frac{d}{dx} [x^n] = nx^{n-1}$	1. relate the average rate of change of a function over an interval to the slope of a secant line in real-world contexts 2. interpret the derivative of a function as both the instantaneous rate of change and the slope of the tangent line to a graph 3. apply appropriate derivative notation

Students learn about	Students should be able to
Function behaviour • increasing/decreasing • intervals where increasing/decreasing • local maxima and minima • absolute maximum and minimum values Applications of differentiation • velocity (v) and acceleration (a) as rates of change: ○ $v = \frac{ds}{dt}$ ○ $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$ • optimising contexts ○ maximising profit ○ minimising cost, loss ○ maximising/minimising area of 2D shapes (circle, quadrilateral) subject to constraints ○ maximising/minimising surface area and volume of 3D solids (cuboid, cube, cylinder, triangular prism, sphere, hemisphere, cone) subject to constraints	4. relate differentiation to the behaviour of the graph of a function 5. use differentiation to sketch the graph of a function 6. use differentiation to justify and interpret proposed solutions to contextualised problems involving optimisation (one function only), equations of tangent lines, velocity and acceleration

Teaching for student learning

Leaving Certificate Mathematics supports a range of teaching and learning approaches that respond to the strengths, needs and interests of all students. The course is student-centred in its design and focuses on students actively doing mathematics rather than only reproducing procedures. In classroom practice, this means teachers might introduce new topics through short contextual problems or questions that prompt students to think, estimate or predict before formal methods are taught. For example, a lesson might begin with a real situation involving data, measurement or change, and students are asked to suggest possible approaches before the underlying technique is developed.

Teachers support learning by planning tasks that require students to work mathematically from the outset. This can include asking students to solve non-routine problems, identify patterns, test conjectures and explain whether answers make sense in context. For example, students might be given a problem with more than one possible method, asked to compare solutions and then justify which approach is most efficient or reliable. Teachers might also design tasks where students model a real-world situation, test assumptions with data, discuss the consequences of different solutions and reflect on how mathematical decisions can affect people, resources and outcomes in practical settings such as finance, planning or data use.

Problem-solving and reasoning are developed through regular classroom routines. Teachers might provide short daily or weekly problem-solving tasks where students are not told which method to use, but instead must select and justify an approach. In supporting proof and justification, teachers might ask students to explain why a rule works in a specific case, check whether a statement holds for different values, or identify and correct errors in a worked solution. Classroom discussion of different methods can be structured so that students compare efficiency, accuracy and clarity of reasoning.

Teachers also support students in interpreting and using mathematical information in different forms. In practice, this might involve students working with graphs, tables, diagrams or digital data sets, and being asked to describe relationships, identify trends and draw conclusions. Teachers might use questioning such as asking students to interpret what a graph implies in a real context, or to explain how changing one variable affects another. Digital tools can be used where appropriate to explore patterns, check results or visualise relationships dynamically.

Collaboration and communication can be embedded through planned classroom structures. Teachers might organise pair or group work where students are assigned roles such as solver, checker or explainer, or where each group presents and defends a solution to the class. Students might be asked to write short explanations of their reasoning, produce annotated diagrams, or present solutions using appropriate mathematical language. Teachers can model and reinforce expectations for clear explanation, precise terminology and logical structure in student responses.

To support inclusion, teachers might adjust the level of challenge within the same task by varying numbers, scaffolding steps, or providing prompts for some students while offering extension questions for others. For example, some students may work from partially completed solutions while others are asked to generalise results or extend a pattern. Teachers might also vary the level of support during independent work, providing targeted guidance where needed and encouraging greater independence where appropriate.

Assessment is used during teaching to inform next steps in learning. Teachers might use quick questioning, short quizzes or review of student work to identify misconceptions and adjust instruction. Feedback is most effective when it directs students to specific improvements, such as reworking a step in a solution, checking assumptions, or improving clarity in a written explanation. Students might also be asked to correct their own work or respond to feedback by attempting a similar problem with improved understanding.

Digital technology

Digital technology can enhance learning, teaching and assessment, creating opportunities for students to develop mathematics knowledge, skills and digital media literacy in ways that cannot be achieved without the use of technology. As students engage with Leaving Certificate Mathematics they have opportunities to use digital technology in a range of ways. For example, they may use digital technology to:

- investigate patterns, test conjectures, model real-world problems and perform complex calculations efficiently
- support multiple representations of mathematical ideas to strengthen understanding, such as moving flexibly between tables, graphs, symbolic expressions and diagrams and using graphing tools to help students compare how the same relationship is represented in different forms

- prepare students for aspects of the AAC, such as using data visualisation tools to compare findings with broader contexts or additional data (for instance, overlaying multiple data samples on a graph), so students can draw conclusions, recognise limitations, and consider how their results might apply beyond the given dataset
- analyse the effect of changing parameters
- perform simulations
- develop a deeper understanding of data through exploration of tools for data collation, visualisation, analysis and representation of results
- become more independent learners through, for example, appropriate online supports
- enhance their experience in the mathematics classroom by engaging in interactive whole-class discussions using shared digital displays, collaboratively exploring problems and testing ideas quickly using technology
- develop their understanding of how mathematicians use digital technology in their work.

Assessment

Assessment in senior cycle involves gathering, interpreting, using and reporting information about the processes and outcomes of learning. It takes different forms and is used for a variety of purposes. It is used to determine the appropriate route for students through a differentiated curriculum, to identify specific areas of strength or difficulty for a given student and to test and certify achievement. Assessment supports and improves learning by helping students and teachers to identify next steps in the teaching and learning process.

As well as varied teaching strategies, varied assessment strategies will support student learning. Feedback throughout the learning process provides information to teachers and students that can be used to modify teaching and learning activities to suit individual learners. By setting appropriate and engaging tasks, asking questions and giving feedback that promotes learner autonomy, assessment will support learning, progression, the development of student key competencies, as well as summarise achievement.

Assessment for certification

Assessment for certification is based on the rationale, aims and learning outcomes of this specification. There are two assessment components: a written examination and an additional assessment component: Investigating with Mathematics. The Investigating with Mathematics task will be based on a brief. Each component will be set and examined by the State Examinations Commission (SEC).

In the written examination, questions will require students to demonstrate learning across the four strands (The unifying strand, strands 1, 2 and 4). The questions will be designed using appropriate stimulus material and structured support.

Table 2 Overview of assessment for certification

Assessment component		Weighting
Investigating with Mathematics	Investigating with Data	30%
	Investigating with Pattern	10%
Written examination		60%

Additional assessment component: Investigating with Mathematics

The Investigating with Mathematics Additional Assessment Component (AAC) provides an opportunity for students to display evidence of their learning throughout the course, in particular, the learning related to

- the problem-solving cycle set out in the unifying strand (U2)
- the statistical enquiry cycle set out in strand 3 and the outcomes (3.1–3.6)
- pattern set out in strand 1 (1.3).

Investigating with Mathematics will be a single-day event, with students separately completing two independent sections: Investigating with Data and Investigating with Pattern in a controlled environment. It is envisaged that the AAC will be completed within a set duration and will take place at the end of fifth year, within the timeframe specified by the SEC. The assessment will be guided by a contextual brief, published by the SEC in advance of the single-day event. This **contextual brief** will:

- set out the specific requirement for the AAC and the procedure for submission
- provide necessary information and context to help students understand the tasks that they will engage with on the day of the assessment
- support teachers in planning to support students' preparation.

Section 1: Investigating with Data

The Investigating with Data task requires students to explore a real-life issue through a statistical lens, using data provided by the State Examinations Commission (SEC). They submit evidence of their ability to engage in the statistical enquiry cycle, from formulating questions and selecting data to analysing relevant data and communicating and reflecting on their findings.

It has been designed to be motivating and relevant for students, to draw on the key competencies and selected learning outcomes developed throughout the course and to highlight the relevance of Mathematics learning to their lives.

As they consider a real-life issue with Mathematics, the Investigating with Data task is an opportunity for students to work like statisticians, to make their own investigative decisions and to develop as statistically and critically aware citizens.

Task brief

On the day that the students are completing the AAC, a **task brief** including a data set will be provided by the SEC. The task brief will:

- include stimulus material to set a context for the data; the targeted population who provided the data and the method in which it was gathered
- allow students to formulate a statistical question related to the issue that they can shed light on, justified with a relevant selection of the data provided

To complete the Investigating with Data task, students follow the statistical enquiry cycle which involves,

- considering an issue to investigate related to the task brief
- generating a relevant statistical question, appropriately selecting and analysing relevant data from the data set provided
- displaying the data in a manner fit for purpose, analysing the data, describing and accounting for variability, interpreting and generalising the results taking variability into account and appropriately communicating the results with reference to the issue being investigated.

Students develop an evidence-based response to their question related to an issue prompted by engagement with the task brief. Upon completion, students submit their work to the SEC.

A separate document, *Guidance to Support the Completion of the Investigating with Mathematics Tasks*, gives guidance on a range of matters related to the organisation, implementation and oversight of the investigative tasks outlined in sections 1 and 2.

Descriptors of quality for the Investigating with Data Task

The descriptors below relate to the learning achieved by students in the Investigating with Data task. In particular, this investigation requires students to:

- consider how data can illuminate a real-world issue
- demonstrate statistical literacy

- relate their statistical investigative work to a relevant societal issue
- communicate their findings appropriately and effectively.

Table 3 Descriptors of quality: Investigating with Data Task

	Students demonstrating a high level of achievement	Students demonstrating a moderate level of achievement	Students demonstrating a low level of achievement
Formulate statistical investigation question	clearly formulate a relevant, focused and statistically investigable question or hypothesis	state a question or hypothesis that is somewhat relevant but lacks focus or statistical clarity	state a vague, irrelevant, or non-statistical question or hypothesis
Selecting and Analysing data	select appropriate, reliable and relevant data, demonstrate critical evaluation of data quality and limitations, effectively organise, summarise and analyse data using appropriate displays and measures of centre and spread	select data with some relevance, perform basic analysis with some appropriate techniques which may contain minor errors or lack depth	use data that is irrelevant, unreliable or lacks any evaluation, shows little or incorrect analysis with inappropriate or no statistical techniques
Interpreting the data	provide clear accurate interpretation of results supported by evidence from the data, making inferences that are relevant and meaningful demonstrating critical thinking and strong awareness of context, clearly explain how the sample is related to the population, recognise limitations of inference	provide a general interpretation of results with some supporting evidence and limited contextual awareness, make some connection between sample and population, but explanation is incomplete or imprecise	misinterpret results or provide little/no explanation linked to the data, make weak or no connection between sample and population, lack awareness of inference implications

Communicating the findings	present findings logically and coherently with well-labelled visuals (charts, graphs). Use appropriate statistical language throughout.	present findings in a structured format, but with some errors in visuals or statistical terminology.	present findings that lack structure; visuals may be missing or unclear; statistical language not used.
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Section 2: Investigating with Pattern

In the Investigating with Pattern task, students engage with a rich mathematical task based on a visual, written or numerical stimulus (provided by the State Examinations Commission) that contains a discernible pattern. They submit evidence to the SEC demonstrating their ability to observe, analyse and interpret this pattern, and to use reasoning and problem-solving skills to explore and communicate how it behaves.

This task is designed to be motivating and accessible for students. It draws on key competencies and selected learning outcomes developed throughout the course, which are essential for building deeper conceptual understanding and for preparing students for more advanced mathematical thinking in sixth year.

As students examine the stimulus and identify key features and repeated structures, the Investigating with Pattern task provides an opportunity for them to work like mathematicians. They pose questions, make and test conjectures, select and apply appropriate strategies, justify their reasoning, generalise their findings and communicate their observations.

Students develop an evidence-based response to the task and submit their work to the SEC.

A separate document, *Guidance to Support the Completion of the Investigating with Mathematics Tasks*, gives guidance on a range of matters related to the organisation, implementation and oversight of the investigative tasks outlined in sections 1 and 2.

Descriptors of quality for the Investigating with Pattern Task

The descriptors below relate to the learning achieved by students in the Investigating with Pattern task. In particular, this investigation requires students to:

- explore and describe mathematical patterns
- make and test conjectures

- apply the problem-solving process
- justify thinking using mathematical reasoning and language.

Table 4 Descriptors of quality: Investigating with Pattern Task

	Students demonstrating a high level of achievement	Students demonstrating a moderate level of achievement	Students demonstrating a low level of achievement
Exploring and analysing the pattern	identify all key features of the visual pattern, describe the pattern's behaviour with precision and use multiple representations (numerical tables, diagrams, algebraic expressions).	identify the main features of the pattern but may overlook some structural elements or relationships, provide some representations (table of values or simple sketches) and attempt to describe the pattern's growth, though explanations may be incomplete or partially general.	describe the pattern superficially or incorrectly; key features or relationships are missed, provide little or no representation of how the pattern changes over stages; may only restate obvious visual observations without deeper analysis.
Mathematical reasoning	develop a well-justified generalisation that is accurate and logically derived from observation, use appropriate mathematical tools (e.g. algebra, functions, sequences, geometric reasoning) to validate the pattern and test conjectures.	suggest a generalisation that is plausible but may contain minor errors or lack full justification, some evidence of testing or validating the idea, but not consistently.	generalisation is incorrect, unsupported or absent and show minimal attempt to justify thinking or test ideas.
Communicating the findings	present a well-organised explanation with clear mathematical language, notation and diagrams, justify each step of the reasoning and demonstrate a strong understanding of why the pattern behaves as it does.	work is understandable and uses mostly correct mathematical language and notation, though some steps in reasoning may be missing or unclear.	explanations are disorganised, incomplete or difficult to follow, mathematical notation, language, or diagrams are incorrect or missing.

Written examination

The written examination will consist of a range of question types. The senior cycle key competencies (figure 2) are embedded in the learning outcomes and will be assessed in the context of the learning outcomes. The written examination paper will include a selection of questions that will assess, appropriate to each level, the learning described in the four strands of study (The unifying strand, strands 1, 2 and 4).

Reasonable accommodations

This Leaving Certificate Mathematics specification requires that students engage with the nature of the subject on an ongoing basis throughout the course. The assessment for certification in Leaving Certificate Mathematics involves a written examination worth 60% of the available marks and an additional component worth 40%. In this context, the scheme of Reasonable Accommodations, operated by the State Examinations Commission (SEC), is designed to assist students who would have difficulty in accessing the examination or communicating what they know to an examiner because of a physical, visual, sensory, hearing, or learning difficulty. The scheme assists such students to demonstrate what they know and can do, without compromising the integrity of the assessment. The focus of the scheme is on removing barriers to access, while retaining the need to assess the same underlying knowledge, skills, values and dispositions as are assessed for all other students and to apply the same standards of achievement as apply to all other students. The Commission makes every effort when implementing this scheme to accommodate individual assessment needs through these accommodations.

There are circumstances in which the requirement to demonstrate certain areas of learning when students are being assessed for certification can be waived or exempted, provided that this does not compromise the overall integrity of the assessment. However, some of the areas of learning in a subject specification cannot be waived because they are core to the subject specification.

More detailed information about the scheme of Reasonable Accommodations in the Certificate Examinations, including the accommodations available and the circumstances in which they may apply, is available from the State Examinations Commission's Reasonable Accommodations Section.

Leaving Certificate Grading

Leaving Certificate Mathematics will be graded using an 8-point grading scale. The highest grade is a Grade 1; the lowest grade is a Grade 8. The highest seven grades (1–7) divide the marks range 100% to 30% into seven equal grade bands 10% wide, with a grade 8 being awarded for percentage marks of less than 30%. The grades at Higher level and Ordinary level are distinguished by prefixing the grade with H or O respectively, giving H1–H8 at Higher level, O1–O8 at Ordinary level.

Table 5 Leaving Certificate Grading

Grade	% marks
H1/O1	90 – 100
H2/O2	80 < 90
H3/O3	70 < 80
H4/O4	60 < 70
H5/O5	50 < 60
H6/O6	40 < 50
H7/O7	30 < 40
H8/O8	< 30

Appendices

Appendix 1: Glossary of action verbs

Action verb	Students should be able to
Analyse	study or examine something in detail, break down in order to bring out the essential elements or structure; identify parts and relationships and to interpret information to reach conclusions
Apply	select and use information and/or knowledge and understanding to explain a given situation or real circumstances
Appreciate	recognise the meaning of, have a practical understanding of
Calculate	obtain a numerical answer, showing the relevant stages in the working
Classify	group things based on common characteristics
Compare	give an account of the similarities and (or) differences between two (or more) items or situations, referring to both (all) of them throughout
Connect/make connections	realise how two or more ideas are related or associated
Consider	take into account
Define	give a precise meaning of a word, phrase, concept or physical quantity
Demonstrate	prove or make clear by reasoning or evidence, illustrating with examples or practical application
Derive	arrive at a statement or formula through a process of logical deduction; manipulate a mathematical relationship to give a requested equation or relationship
Describe	develop a detailed picture or image of, for example, a structure or a process, using words or diagrams where appropriate; produce a plan, simulation, or model

Action verb	Students should be able to
Determine	obtain the only possible answer by calculation or by substituting measured or known values of other quantities into a standard formula
Estimate	give a reasoned order of magnitude statement or calculation of a quantity
Evaluate (data)	collect and examine data to make judgments and appraisals; describe how evidence supports or does not support a conclusion in an inquiry or investigation; identify the limitations of data in conclusions; make judgments about the ideas, solutions or methods
Evaluate (ethical judgement)	collect and examine evidence to make judgments and appraisals; describe how evidence supports or does not support a judgement; identify the limitations of evidence in conclusions; make judgments about the ideas, solutions or methods
Explain	give a detailed account including reasons or causes
Examine	consider an argument or concept in a way that uncovers the assumptions and relationships of the issue
Explore	observe, or study in order to establish facts
Find	discover by study or experiment
Formulate	to reduce to or express in a formula
Generalise	generate a general statement based on specific instances
Generate	produce or create
Identify	recognise patterns, facts, or details; provide an answer from a number of possibilities; recognise and state briefly a distinguishing fact or feature
Illustrate	use examples to describe something
Investigate	observe, study, or make a detailed and systematic examination, in order to establish facts and reach new conclusions

Action verb	Students should be able to
Justify	give valid reasons or evidence to support an answer or conclusion
Manipulate	skillfully transform and rearrange algebraic expressions using their structure and mathematical rules while preserving their equivalence
Measure	quantify changes in systems by reading a measuring tool
Model	make justified predictions, describe phenomena and solve problems using words/diagrams/numbers/graphs/equations as appropriate
Perform	carry out in a formal manner
Pose	put forward for consideration
Predict	give an expected result of an event; explain a new event based on observations or information using logical connections between pieces of information
Prove	establish the truth of a mathematical statement through a sequence of logical mathematical arguments
Provide evidence	provide data and documentation that support inferences or conclusions
Recognise	identify facts, characteristics, or concepts that are critical (relevant/appropriate) to the understanding of a situation, event, process, or phenomenon
Relate	associate, giving reasons
Represent	to be something or to be equal to something
Select	carefully choose the best or most suitable option
Sketch	make a rough drawing representing the chief features
Solve	to find the answer to a problem
Translate	to change something into a different form
Use	apply knowledge or rules to put theory into practice
Verify	confirm the truth, correctness or reasonableness of a mathematical

Action verb	Students should be able to
	statement, equation or proposed solution by using logical reasoning, substitution, calculation or checking relevant cases

